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To Whom it May Concern:

Sandia National Laboratories (Sandia) is pleased to participate with Prof. Mark Sussman (Department of Mathematics at Florida State University) in their response to DOE's Request for Applications "The Genesis Mission: Transforming Science and Energy with AI" (DE-FOA-0003612) for topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies." The title of the proposal is "Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions."

Mark Sussman has requested Sandia's unique capabilities to be included in their response to this Request for Applications. Subject to available funding and resources, Sandia anticipates participating in mutually agreed tasks, which will be performed by staff in the Thermal/Fluids Science and Engineering Department. The objective will be to develop AI-enhanced sharp-interface numerical methods that can reduce the computational cost in R&D of advanced combustion devices (such as jet engines and rotating detonation rockets). This activity will take place using existing DOE-developed simulation codes. Sandia staff will provide critical domain knowledge to anchor the choice of the user cases, while also assisting in the setup and postprocessing of the simulations. Crucially, the staff will test the AI-augmented workflow in comparison with existing best practices.

Sandia National Laboratories is a government-owned, contractor-operated facility operated by National Technology & Engineering Solutions of Sandia, LLC (NTESS) for the U.S. Department of Energy (DOE)/ National Nuclear Security Administration (NNSA) under Management and Operating (M&O) Contract DE-NA0003525. As a DOE M&O Contractor, NTESS is obligated to offer its unique services to entities that wish to acquire these services using a DOE-approved agreement mechanism. Providing Sandia's services to Prof. Mark Sussman is subject to DOE review and approval in accordance with DOE policies and regulations. These services may be acquired under a DOE-approved bilateral contract with Sandia post award. Sandia implements internal training that aligns with research security training guidelines and is consistent with 42 USC 19234 and DOE Order 486.1A.

This information is provided in accordance with applicable DOE policy regarding Sandia's participation in federal government announcements. The Sandia point of contact for this potential collaboration is Marco Arienti, Sandia National Laboratories, P.O. Box 969, Livermore CA 94551.

Sincerely,
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